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Long-timescale Plasticity Mediated by ER-Dependent Synaptic Stabilization

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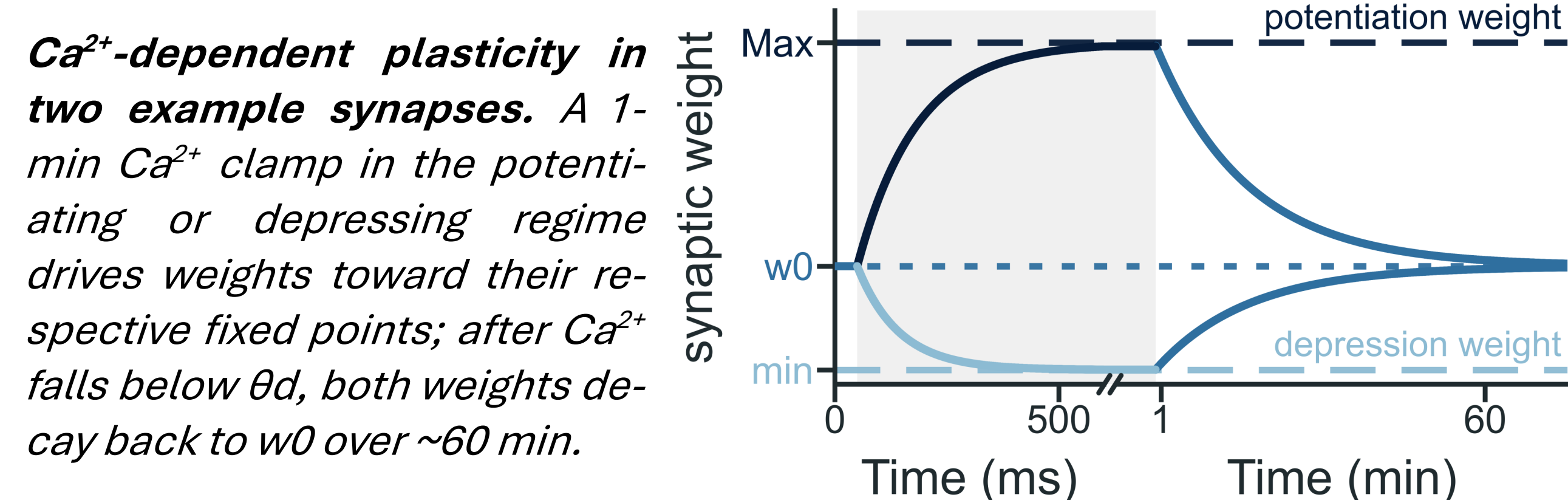
Introduction

- Synaptic plasticity involves multiple timescales: fast Ca^{2+} dependent induction over milliseconds-to-seconds¹ and slow maintenance across minutes-to-hours².
- It remains unknown whether any single biophysical phenomenon can account for multiple timescales of plasticity.
- The endoplasmic reticulum (ER) is a viable candidate, as it enters dendritic spines on the order of seconds³ and enables spine-specific plasticity maintenance over hours⁴.
- **By jointly modeling ER-trafficking and its stabilizing effects on Ca^{2+} plasticity, we show that it may account for network plasticity over longer timescales.**

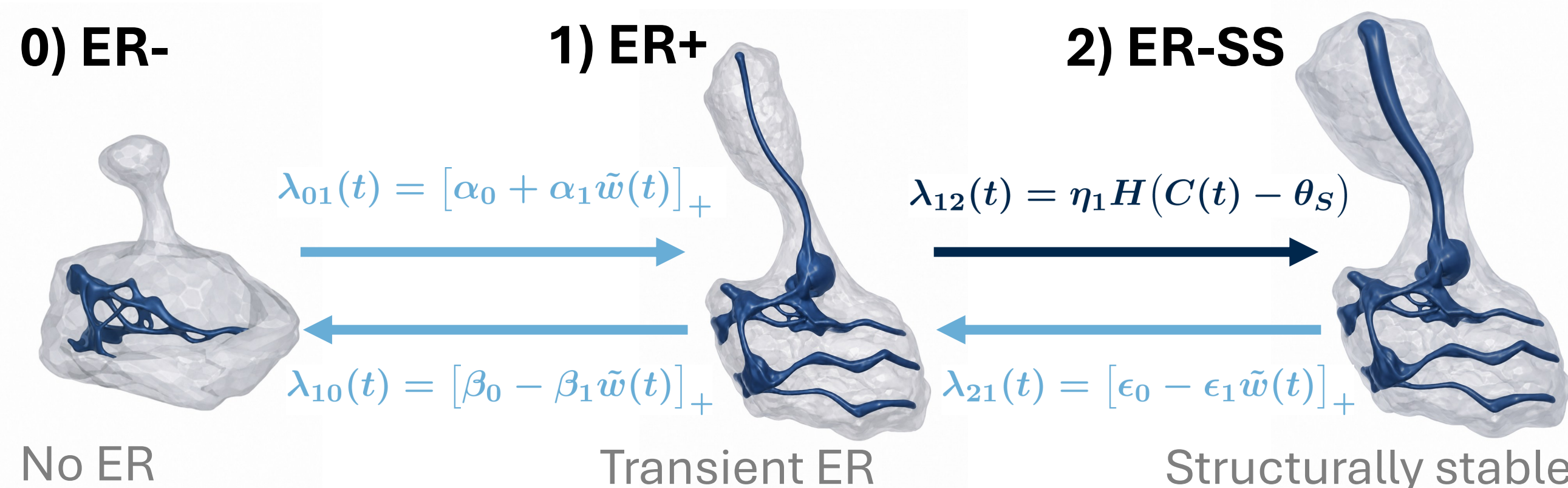
Methods

Phenomenological Ca^{2+} plasticity

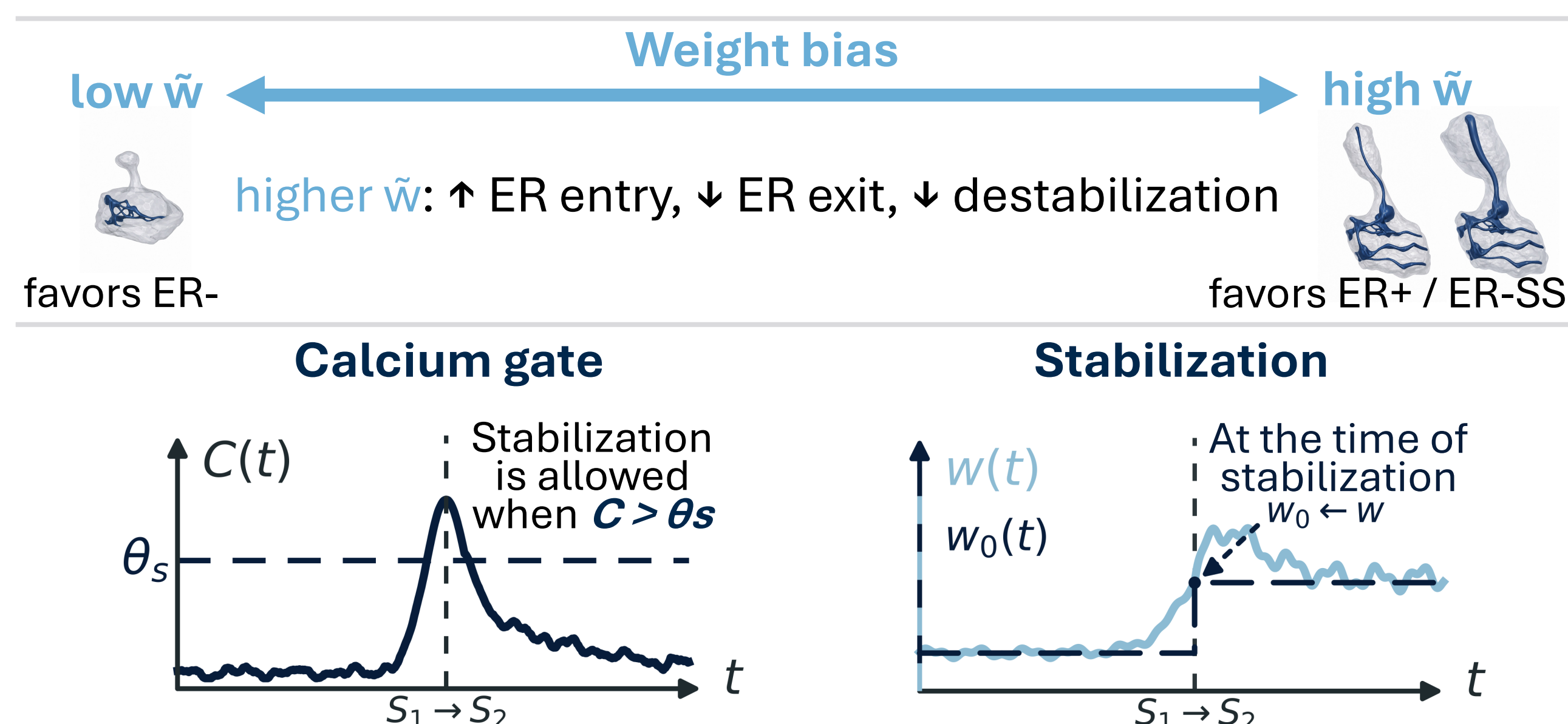
- We use an established plasticity rule⁵ in which each synapse has an activity-dependent Ca^{2+} trace, $C(t)$.
- Weights potentiate when $C(t) > \theta p$, depress when $\theta d < C(t) < \theta p$, and slowly return to a baseline weight, w_0 , when $C(t) < \theta d$. w_0 is a novel feature added to capture post-potential spine shrinkage observed without spine ER⁴.



Markov model of spine ER dynamics



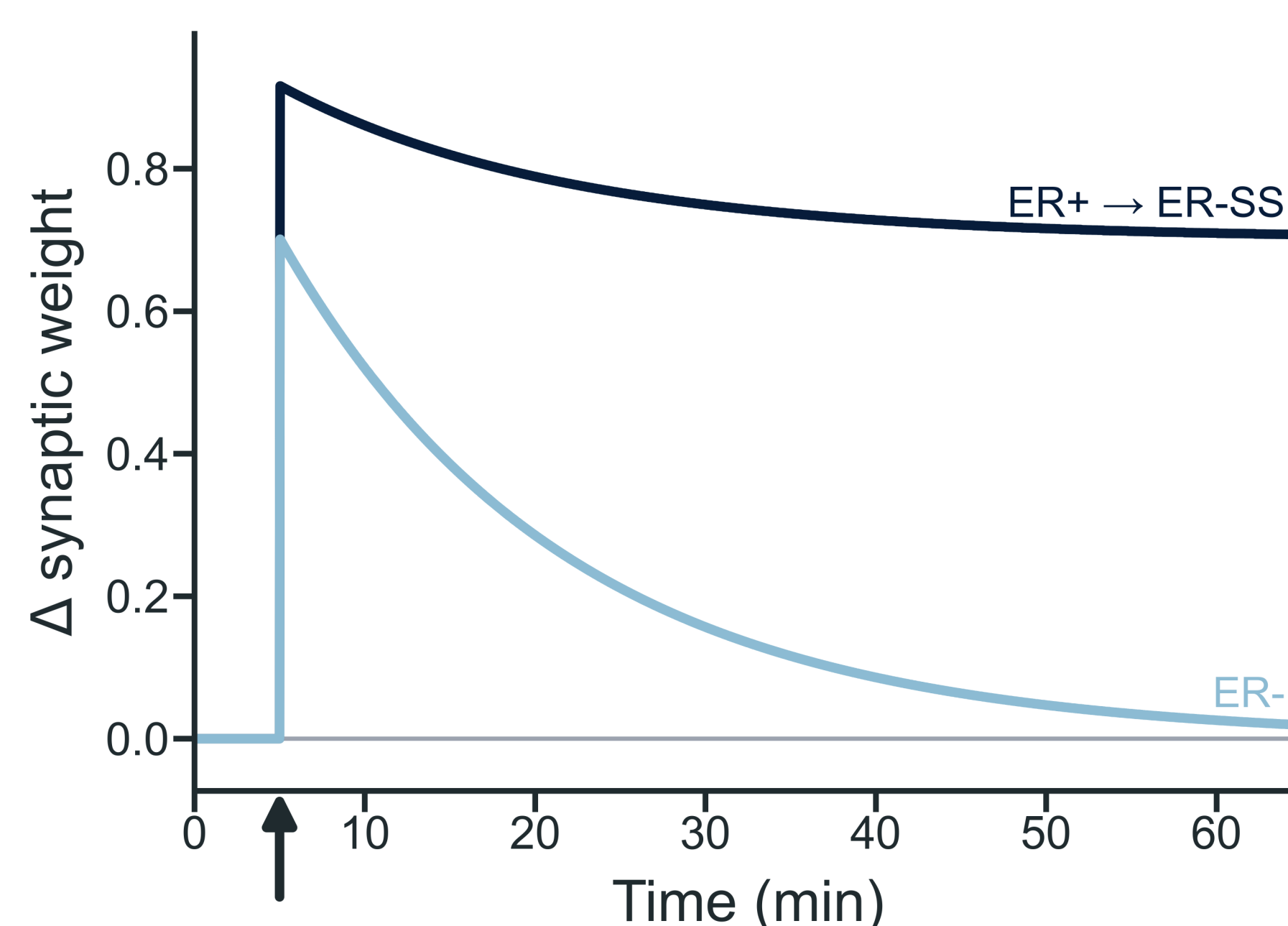
Description of ER model variables. Rate parameters: α , entry; β , exit; η , Ca^{2+} dependent stabilization; ϵ , destabilization. $\tilde{w}$ is the difference from the reference weight, w_0 , normalized by the maximum weight. H : Heaviside step function.



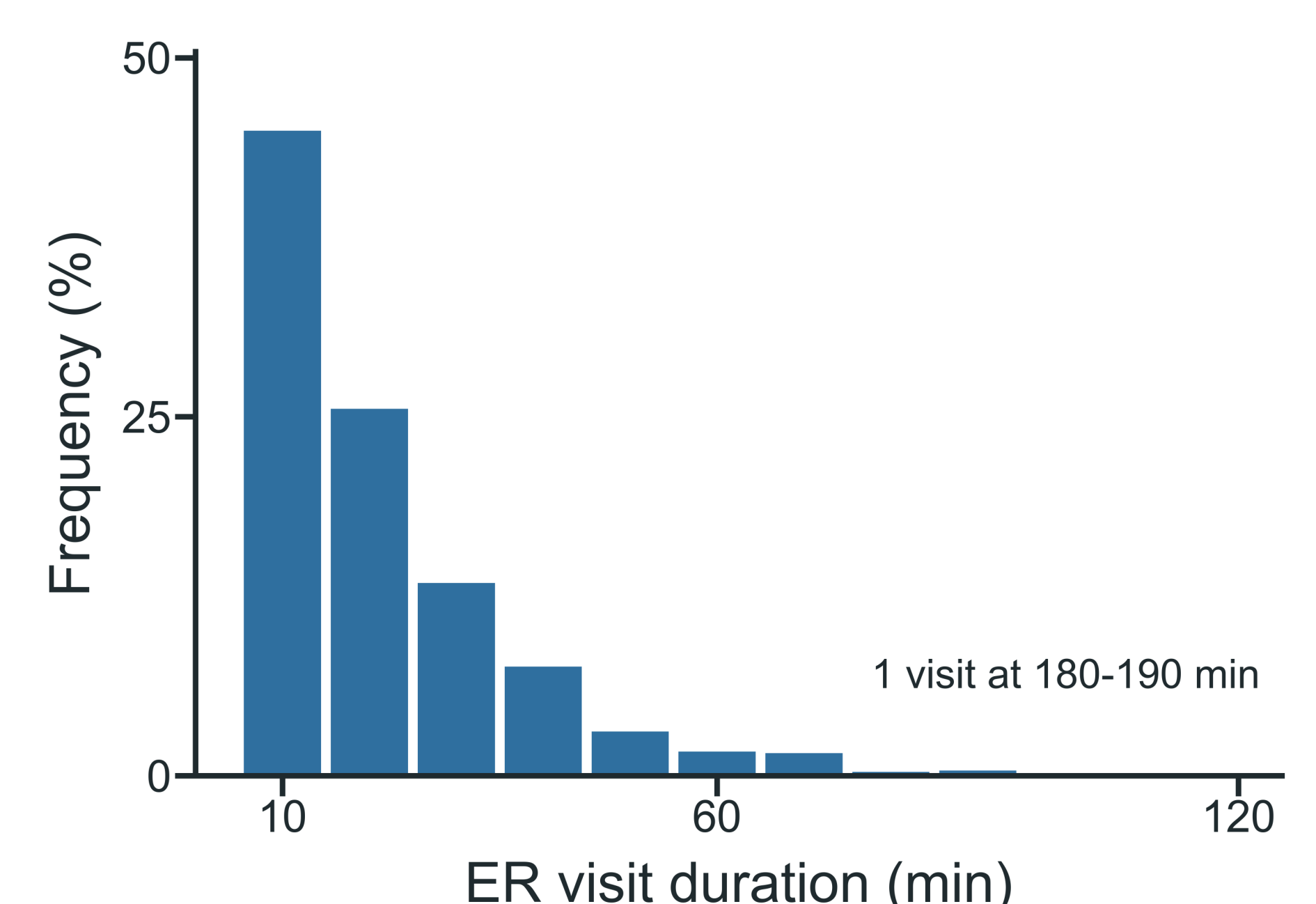
Conclusions & future directions

- **ER-dependent maintenance of Ca^{2+} plasticity allows synaptic changes to persist in E/I RNNs on supra-minute timescales.**
- This model of long timescale plasticity is uniquely grounded by an experimentally observed biological mechanism, enabling us to address theoretical questions about learning & memory previously posed by more abstract models⁷.
- Future work will study how ER dynamics effect learning & memory in spiking networks based on the formation and maintenance of neuronal assemblies.

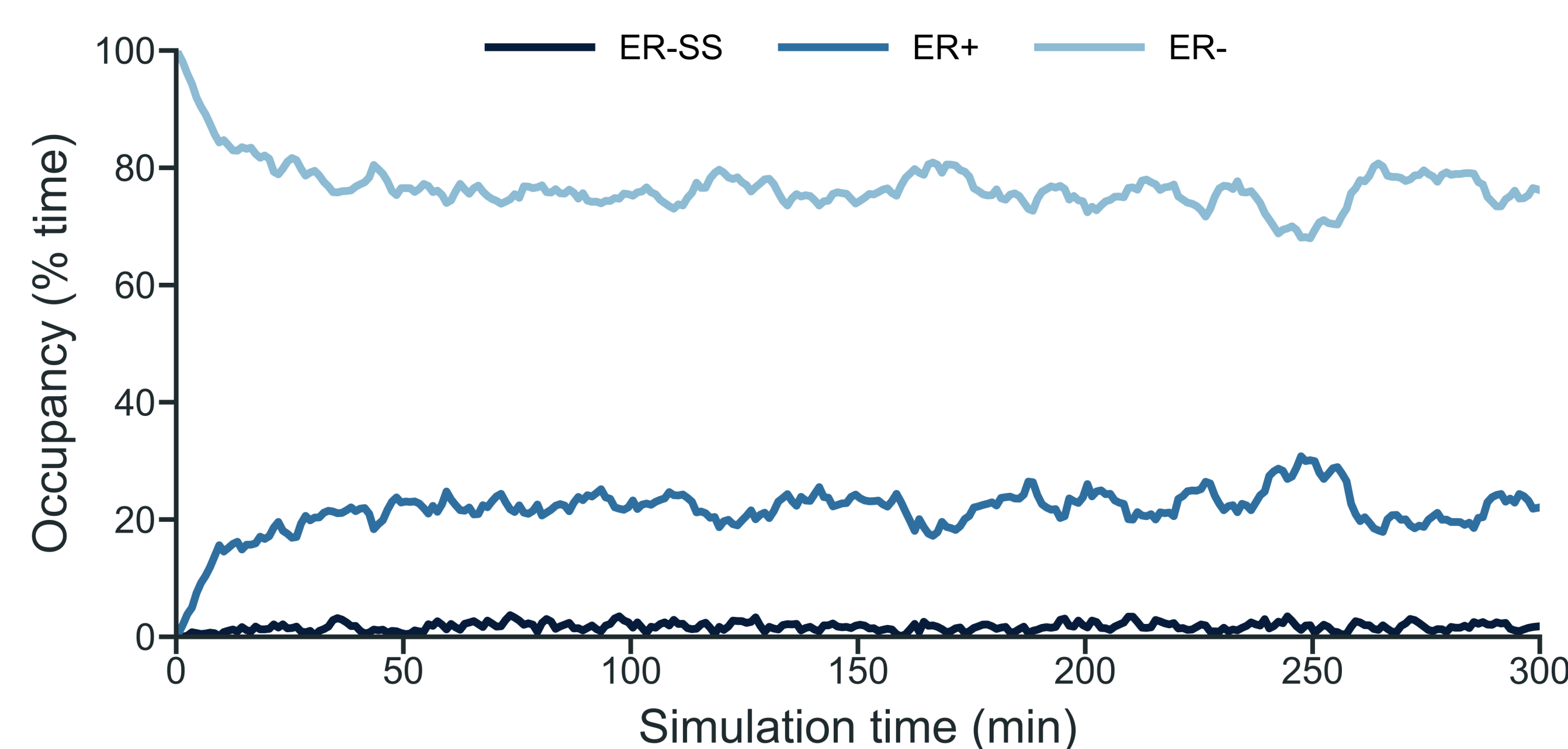
Markovian ER dynamics capture known trafficking and plasticity related phenomena in single spines



Differential response of ER- and ER+ spines to an potentiating stimulus. The arrow represents 100Hz coincident pre- and post-synaptic spiking for 1 s. The ER+ spine transitions to ER-SS and maintains its potentiated weight, as seen in vitro⁴.

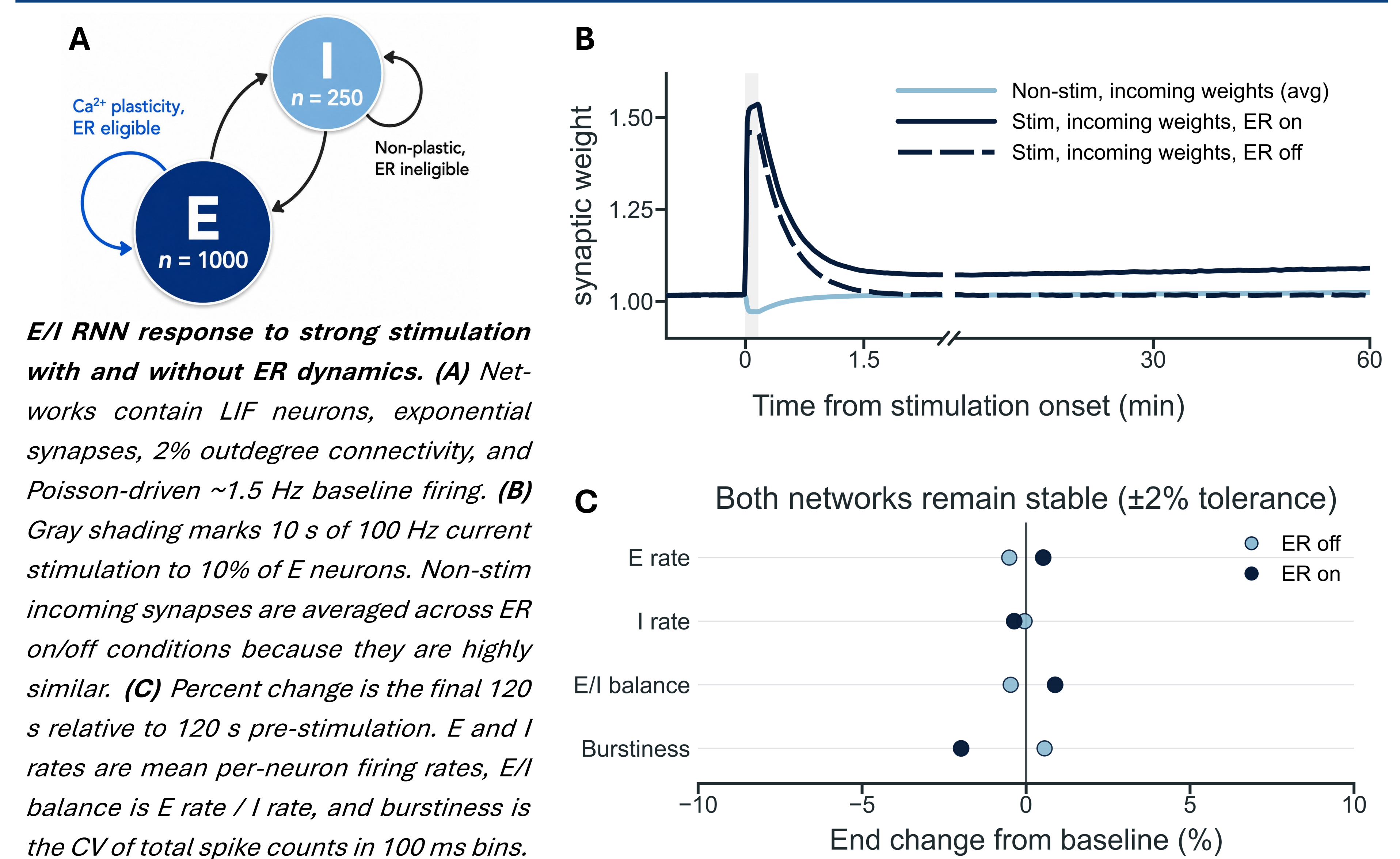


Distribution of ER visit durations is right-skewed. Under 1.5Hz Poisson spiking, most ER visits are short (<10min), longer visits (>60min) are rare, and the distribution is heavy-tailed, as seen in in vitro³. $n(\text{synapses})=250$, $n(\text{visits})=1,064$.



ER occupancy is stable under noise. Occupancy is the % of spines in a state at any time. Synapses receive spike times for pre- and post-synaptic neurons controlled by a 1.5Hz Poisson process. ~20% of spines are ER+ at any given time, as seen in vivo⁶ & in vitro³. $n(\text{synapses})=250$.

ER-dependent stabilization preserves potentiated synapses without disrupting E/I network stability



Citations

1. Lynch *et al.* Intracellular injections of EGTA block induction of hippocampal long-term potentiation. *Nature* (1983).
2. Frey *et al.* Anisomycin, an inhibitor of protein synthesis, blocks late phases of LTP phenomena in the hippocampal CA1 region in vitro. *Brain Res.* (1988).
3. Perez-Alvarez *et al.* Endoplasmic reticulum visits highly active spines and prevents runaway potentiation of synapses. *Nat. Commun.* (2020).
4. Dittmer *et al.* L-type Ca^{2+} channel activation of STIM1-Orai1 signaling remodels the dendritic spine ER to maintain long-term structural plasticity. *Proc. Natl. Acad. Sci.* (2024).
5. Moldwin *et al.* A generalized mathematical framework for the calcium control hypothesis describes weight-dependent synaptic plasticity. *J. Comput. Neurosci.* (2025).
6. Spacek *et al.* 3D organization of smooth endoplasmic reticulum in hippocampal CA1 dendrites and dendritic spines of the immature and mature rat. *J. Neurosci.* (1997).
7. Fusi *et al.* Cascade models of synaptically stored memories. *Neuron* (2005).